



Tourism and the Economy in Saudi Arabia

This project analyzes tourism and climate data in Saudi Arabia to understand tourism trends and their impact on the economy. The results help support decision-making and tourism development.

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Lab 1 – Project Scenario & Data Sources

Project Scenario & Data Sources

We selected these datasets because they support our project on tourism and the economy in Saudi Arabia. Each dataset provides different insights, including tourism trends, visitor statistics, climate factors, and source markets. Together, they help us better understand the tourism sector and its economic impact

Source Name	Type	Format	Record
Saudi Arabia Tourism & Climate Data (2018–2023)	csv	structured	30850
Arabia Tourism Saudi Dataset (2015-2024)	csv	structured	1058
Top ten inbound source markets	excel	structured	111



Lab 2 – Structured vs. Unstructured Data

Unstructured Data Analysis Table

Attribute	Data Type	Structured/unstructured	Justification
Service ID	integer	Structured	Fixed format unique identifier
Service title EN	Text	structured	
Service title AR	Text	structured	
Service description EN	Text	unstructured	
Service description AR	Text	unstructured	
Service URL	link	structured	

Transformation Plan Table

Step	Technique	Expected structured output
1	Data cleaning(remove special characters,normalize text)	Cleaned service description text
2	Text preprocessing(tokenization,stop-word removal)	Processed keywords
3	Keyword extraction	Identified service-related terms
4	Text classification	Service category (license,renewal,issue,inquiry)
5	Entity extraction	Target group (individual,business,investor)
6	Standardization	Unified category and target group values
7	Schema mapping & loading	Structured table ready for data warehouse

Proposed Structured Output Table

Field Name	Data Type	Source	Description
Service_ID	INT	Unique identifier of the service	
Service_Name_AR	VARCHAR(255)	Service name in Arabic	
Service_Name_EN	VARCHAR(255)	Service name in English	
Service_Category	VARCHAR(100)	Classified type(license,renewal,issue,inquiry)	
Target_Group	VARCHAR(100)	Intended beneficiary(individual,business,investor)	
Service_Objective	TEXT	Main purpose of the service	
Service_Requirements	TEXT	Key conditions or requirements	
Service_URL	VARCHAR(500)	Official service link	



Lab 3 – Data Cleaning & Preprocessing

The data quality assessment identified a few issues across the three tourism datasets. The main problems included inconsistent date formats, duplicate records, missing temperature values, text-formatted numeric data, and inconsistent naming conventions. Data profiling showed that most attributes had no missing values, although some fields contained duplicates and mixed-case values.

Cleaning Strategy

Issue	Cleaning Technique	Rule Applied	Expected Outcome
Date Consistency	Standardization	Convert dates to YYYY-MM	Consistent timeline
Naming Issues	Mapping	Standardize province names	Unified categories
Duplicates	Remove Duplicates	Drop duplicate records	Accurate counts
Missing Values	Imputation	Fill using averages	Complete dataset
Numeric Strings	Parsing	Convert text to numeric	Ready for analysis
Consistent aggregation	Use ISO country codes	Standardization	Country Names
Better distribution	Replace extreme values	Statistical Trimming	Outliers



Lab 4 – Schema Matching

Source A Attribute	Source B Attribute	Match Type	Confidence	Justification
year	YEARS	Partial Match	Medium	Name similarity the difference is only in letter case; both represent the same concept (year).
trips	Tourists Number (Overnight Visitors) (000)	No Match	Low	Semantic mismatch “trips” refers to the number of journeys, while the other attribute represents the number of tourists.
nights	Overnight Stay (000)	Partial Match	Medium	Semantic meaning both refer to the number of nights or stays, expressed differently.
spendSAR	Tourists Spending (SAR Mn)	Partial Match	Medium	Name similarity & semantic meaning both relate to tourist spending in SAR, with slight naming differences.
destinationProvinceNameEn	DESTINATION	Partial Match	Medium	Semantic meaning both refer to the destination, but the first is more specific (province name).



Lab 5 – Schema Mapping

Schema mapping

Source Attribute(s) From source A	Target Attribute(s) From source B	Mapping Type	Short Description
year	YEARS	1 : 1	Same semantic meaning different
nights	Overnight Stay (0 0 0)	1 : 1	Seem semantic different column
spendSAR	Tourists Spending (SAR Mn)	1 : 1	Seem semantic on column different
destinationProvinceNameEn	DESTINATION	1 : 1	Seem semantic on column different

mapping rules

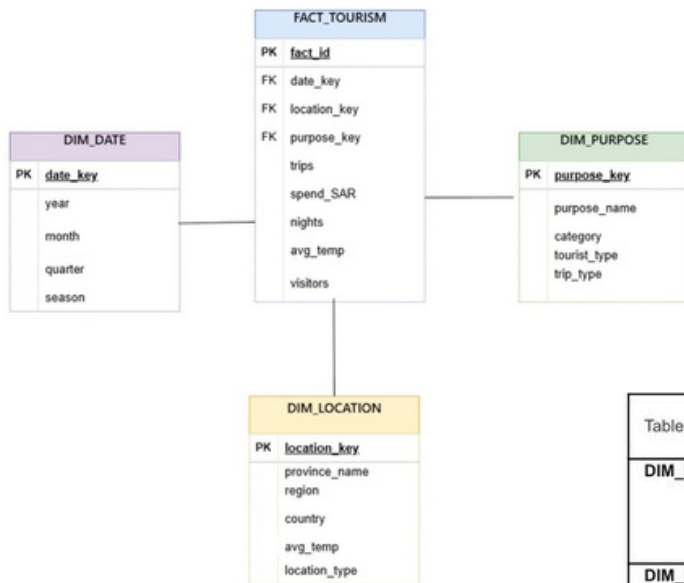
Rule #	Rule Expression	Explanation
1	year=YEAR	Semantic Identity: Both fields represent the same time period (Year)
2	nights= Overnight Stay (0 0 0)	Semantic Equivalence: Both attributes refer to the duration of the stay.
3	spendSAR= Tourists Spending (SAR Mn)	Semantic Equivalence: Both fields represent the total expenditure value.
4	destinationProvinceNameEn= DESTINATION	Semantic Match: Both attributes represent the geographical location.

SQL-style\
SELECT
 Year **AS** YEAR,
 nights **AS** Overnight Stay (0 0 0),
 spendSAR **AS** Tourists Spending (SAR Mn),
 destinationProvinceNameEn **AS** DESTINATION,
FROM source A ;



Lab 6 – Star Schema, Data Warehouse & ETL

Star Schema



Data Warehouse

Table Name	Type	Rows Inserted	Source(s) Used	Notes / Simplifications
DIM_DATE	Dimension	8	tourism_with_temps.csv — year & month columns	Covers Q1–Q4 for years 2022 and 2023; season assigned by quarter
DIM_LOCATION	Dimension	8	tourism_with_temps.csv — destinationProvinceNameEn + ;destination_temp Top_ten_inbound_source_markets_CSV.csv — inbound country	Saudi 7 domestic provinces + 1 inbound country (Kuwait); avg_temp from destination_temp column
DIM_PURPOSE	Dimension	5	tourism_with_temps.csv — visitPurposeEn column	All 5 original categories used: Leisure, Business, VFR, Religious, Other
FACT_TOURISM	Fact	10	tourism_with_temps.csv — subset: month=7, year=2022, destination=Albaha/Aseer/Makkah/Riyadh	trips, spendSAR, nights taken directly from source; visitors = rounded trips value; avg_temp = destination_temp

Extract:

- tourism_with_temps.csv
- Arabia Tourism Saudi Dataset (2015-2024)
- Top_ten_inbound_source_markets.csv

Load:

Transform:

- Filtered data: month=7, year=2022
- visitors = rounded trips value
- Season assigned by quarter
- avg_temp taken from destination_temp column

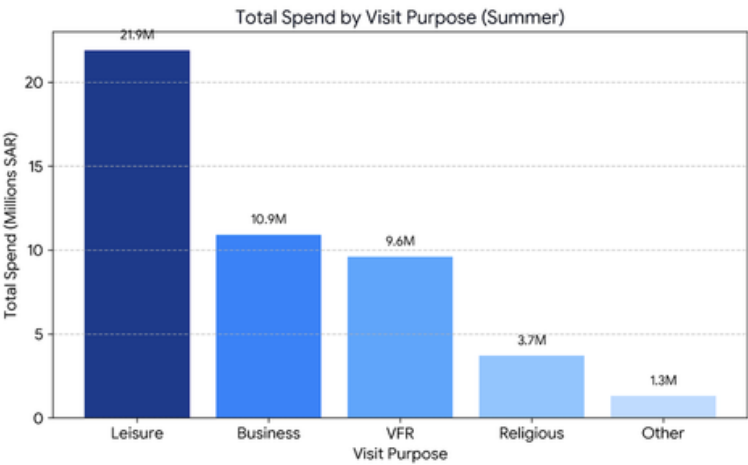
- DIM_DATE → 8 rows
- DIM_LOCATION → 8 rows
- DIM_PURPOSE → 5 rows
- FACT_TOURISM → 10 rows



Lab 7 - Strategic Questions + OLAP/Data Mining

Strategic Business Questions

- Q1: Which visit purposes generate the highest spending across seasons?
- Q2: Is there a correlation between avg temperature and nights stayed?
- Q3: Which regions attract the highest visitors per category?
- Q4: What is the avg spend per trip: domestic vs international?



OLAP Queries & Results

Query	Result
Spending by Purpose & Season	Leisure = highest (21.9M SAR)
Temperature vs Nights	Moderate climate = more nights (Albaha 121K)
Region & Category	Southern dominates recreation
Avg Spend per Trip	Domestic 1,544 SAR vs International 527 SAR



Key Insights + Visual Dashboards or interactive reports

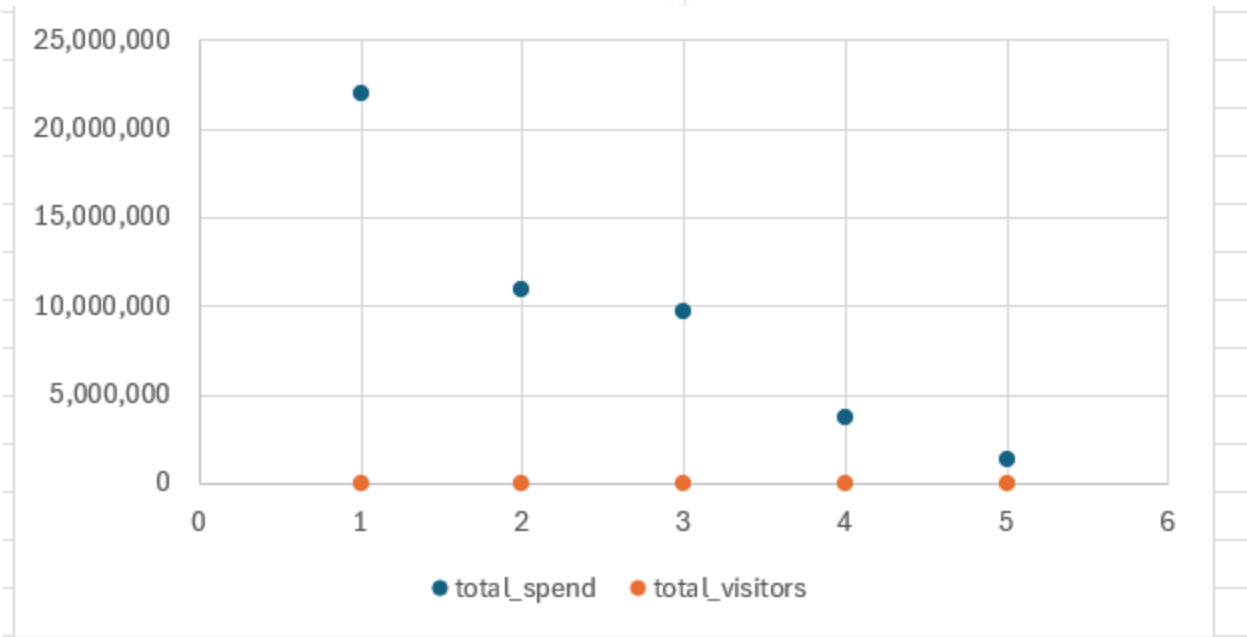
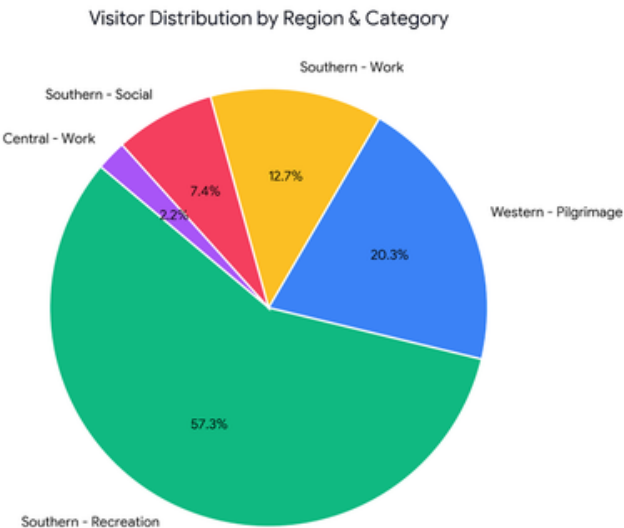
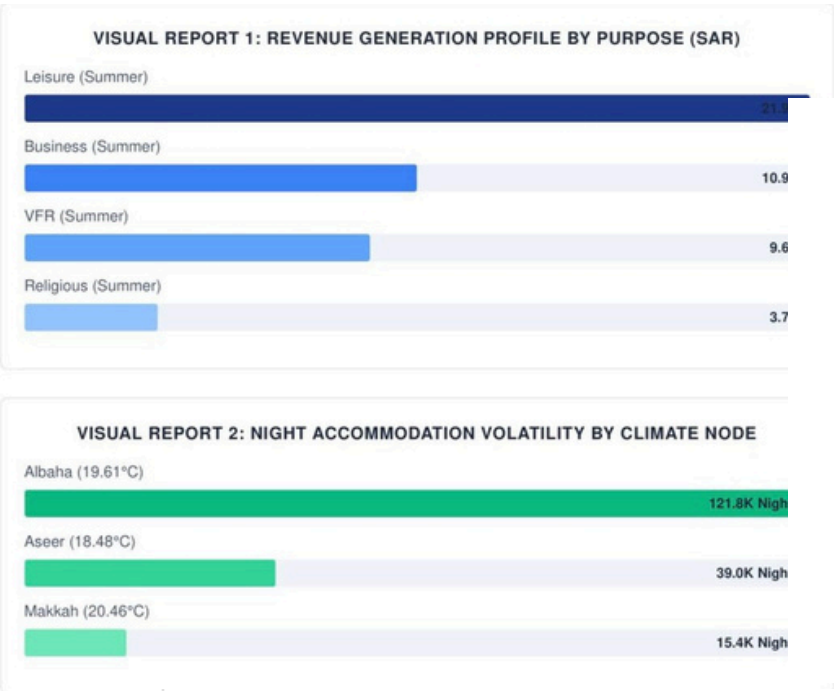
Key Insights

Leisure Tourism generated the highest revenue

Summer is the strongest tourism season

Albaha recorded the highest number of tourist nights

Domestic tourists spend more per trip



Final Recommendations + Lessons Learned

Recommendations

- .Improve data quality
- .Collect more detailed tourism information
- .Update datasets regularly

Lessons Learned

- .Importance of data cleaning
- .Challenges of integrating multiple datasets
- .Benefits of data warehousing

